Fiscal Year:	FY 2005	**Task Last Updated:**	FY 12/13/2005
PI Name:	Martindale, Mark Q Ph.D.		
Project Title:	The Evolution of Gravity Reception in the Metazoa		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Neurobiology		
Space Biology Special Category:	None		
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Zip Code:	96813	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2000 Space Biology 00-OBPR-01
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No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	4
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	2		
COI Name (Institution):	Jacobs, Dave (U.C.L.A.)		
Grant/Contract No.:	NAG 2-1519		
Performance Goal No.:			
Performance Goal Text:			
	We will investigate the evolutionary origins of gravity detection in multicellular organisms. The most basal metazoans capable of gravity reception are members of the Cnidaria and Ctenophora. The apical organ of ctenophores and the rhopalia of pelagic medusoid cnidarians are composed of modified ciliary structures and share many structural similarities to the gravity detectors of higher animals, such as vertebrate vestibular hair cells. In order to determine the possible evolutionary links between gravity amplifiers in the history of life on earth, we intend to study representatives from these two pivotal phyla. We have isolated several genes (e.g. forkhead and hairy/enhancer of split orthologs in ctenophores and brain/POU genes in both ctenophores and cnidarians) that are expressed in the developing sense organs and will continue to isolate candidate genes by PCR and library screening involved in gravity detection. Specific targets are Otx, atonal, bHLH		

Task Description:	genes, Pax, and notch orthologs, all of which are involved in sensory cell development in other systems. The spatial and temporal expression of these genes will be determined by in situ hybridization techniques. Reagents and cell specific markers will be developed to allow future physiological studies of the gravity detection apparatus. Because many of these genes are also expressed in other sensory structures our experiments will also shed light on to the evolutionary relationship between gravity detection and other sensory modalities such as 'hearing' and the ability to detect light. Our work will not only use rigorous phylogenetic reconstruction techniques to help understand the evolution of gravity detection in the Metazoa but will also determine the phylogenetic relationships of these basal metazoan groups to bilaterian forms. The complimentary backgrounds of the two Co-P.I.s, the experience with the development and anatomy of the study organisms, in addition to the preliminary data from both labs, insures that rapid progress on these topics will be made.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This is the first systematic investigation of the evolution of sensory strucutres involved in detecting position in the earth's gravitational field. Our experimental design is intended to determine ancestral and derived aspects of this process and gain a better idea of the developmental and molecular potentialities of ancestral genomic programs.
Task Progress:	Progress from the Martindale Lab We have made progress in isolating potential neural developmental regulatory genes in two species (the starlet sea anemone, Nematostella vectensis, and the ctenophore, Mnemiopsis leidyi) over the past year. Progress with N. vectensis has been facilitated by the sequencing of the genome by the Joint Genome Institute (Dept. Energy). We have isolated and characterized the expression of the following genes in N. vectensis: Otx1, 2, and 3, noggin1, and noggin2, goosecoid, BMP5-8, chordin, follistatin, notch, hairy, and netrin. All of these genes are currently being prepared for publication. The upshot of these experiments is that most of thes genes are expressed in the chemosensory structure called the apical tuft, but they are also, unexpectedly, expressed in discreet regions in the pharynx. We hypothesize that these genes are generating neural precursor in both of these locations and are currently pursuing the characterization of markers for the adult nervous system. It is possible that the pharynx represents the precursor of the bilaterian ‘brain’ and that these genes were co-opted from the apical tuft in a swimming dispersal phase of the cnidarian life history. Progress in the ctenophore embryo has come slower due to its seasonality and the lack of genomic resources. Genome sequencing is still on review at the J.G.I. and we are hopeful that we will be able to obtain sequence in the next year or two. We have isolated two BarX homeodomain-containing genes, two paired class homeodomian containing genes, 4 additional T-box transcription factors (including a brachyury ortholog), and an additional sine occulis gene. We are currently performing in situ hybridizations with these genes and hoe to have publications onthem by the spring of 2006. Progress from the Jacobs Lab: Over the Course of the Grant we recovered markers for sense organ genes from the Jellyfish Aurelia aurita and the Anenome Nematostella vectens. As well as from other basal metazoans, including acoel flatworms and sponges. Genes recovered include members of the POU, Six/sine oculis, LIM, Otx, atonal and eyes absent gene families. Expression work has proceeded with these markers. In addition we have done considerable work involving the neural transmitters GnRH and FMRF as well as markers for tubulin and actin to examine the context of sense organ development in Aurelia. The approach has involved confocal microscopy as well as a combination of florescence microscopy and DIC. These results give us a detailed understanding at the cellular level of the neural and sensory organization of Aurelia during the development of the planula, polyp and early medusa (ephyra). This combined with in situ hybridization and RTPCR expression studies using probes from the genes recovered above provided us with a comprehensive view of the involvement of regulatory genes in the neural and sense organ development in the Jellyfish (Scyphozoa) Aurelia as well as comparative studies on the anemone Nematostella. Our publications relating to this work, as well as abstracts and related scientific presentations are enumerated below. In addition we anticipate the submission of four additional papers in the next 6 to 8 months. These includes studies of the neural markers through all developmental stages, studies of the sensory neural organization of the polyp tentacles as well as at least two additional papers on the expression of developmental markers, including POU and Six genes as well as eyes absent in the development of the sensory structures of the, medusa in Aurelia as well as observations on the polyp and planula in Aurelia and Nematostella.
Bibliography Type:	Description: (Last Updated: 12/26/2007)
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Abstracts for Journals and Proceedings	Jacobs DK, Hughes NC, Winchell CJ. "Terminal Addition and the Evolution of Bilaterian Form." SICB 2005 annual meeting, San Diego, January 2005. ICB 2005 proceedings, 2005 January. , Jan-2005
Abstracts for Journals and Proceedings	Nakanishi N, Camarra A, Jacobs DK. "Evolution of sensory genes in basal Metazoa. " SICB 2005 annual meeting, San Diego, January 2005. ICB proceedings, 2005 January. , Jan-2005
Articles in Peer-reviewed Journals	Matus DQ, Thomsen GH, Martindale MQ. "Dorso/ventral genes are asymmetrically expressed and involved in germ-layer demarcation during cnidarian gastrulation." Curr Biol. 2006 Mar 7;16(5):499-505. PMID: 16527745 , Mar-2006
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